

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715

For ML researchers, AI security practitioners, and cloud infrastructure engineers

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

Providers charge for specific models (e.g. Llama-3-70B, Qwen2-72B) but the black-box nature of APIs means users have no guarantee the advertised model is actually being served. Strong economic incentives drive covert substitution with quantized, smaller, or different models.

1. Smaller Model

e.g. 8B served instead of 70B

2. Quantized Variant

FP8/INT8 instead of FP16

3. Fine-tuned Version

Updated or altered weights

4. Different Family

Completely different architecture

Formal Hypothesis Testing Framework:

H_0 : API serves $P_{\text{spec}}(y|x)$ | H_1 : API serves $P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$

Goal: Given samples from the API, can an auditor distinguish the specified model's distribution from an alternative?

FP8 Quantization Preserves Most Accuracy

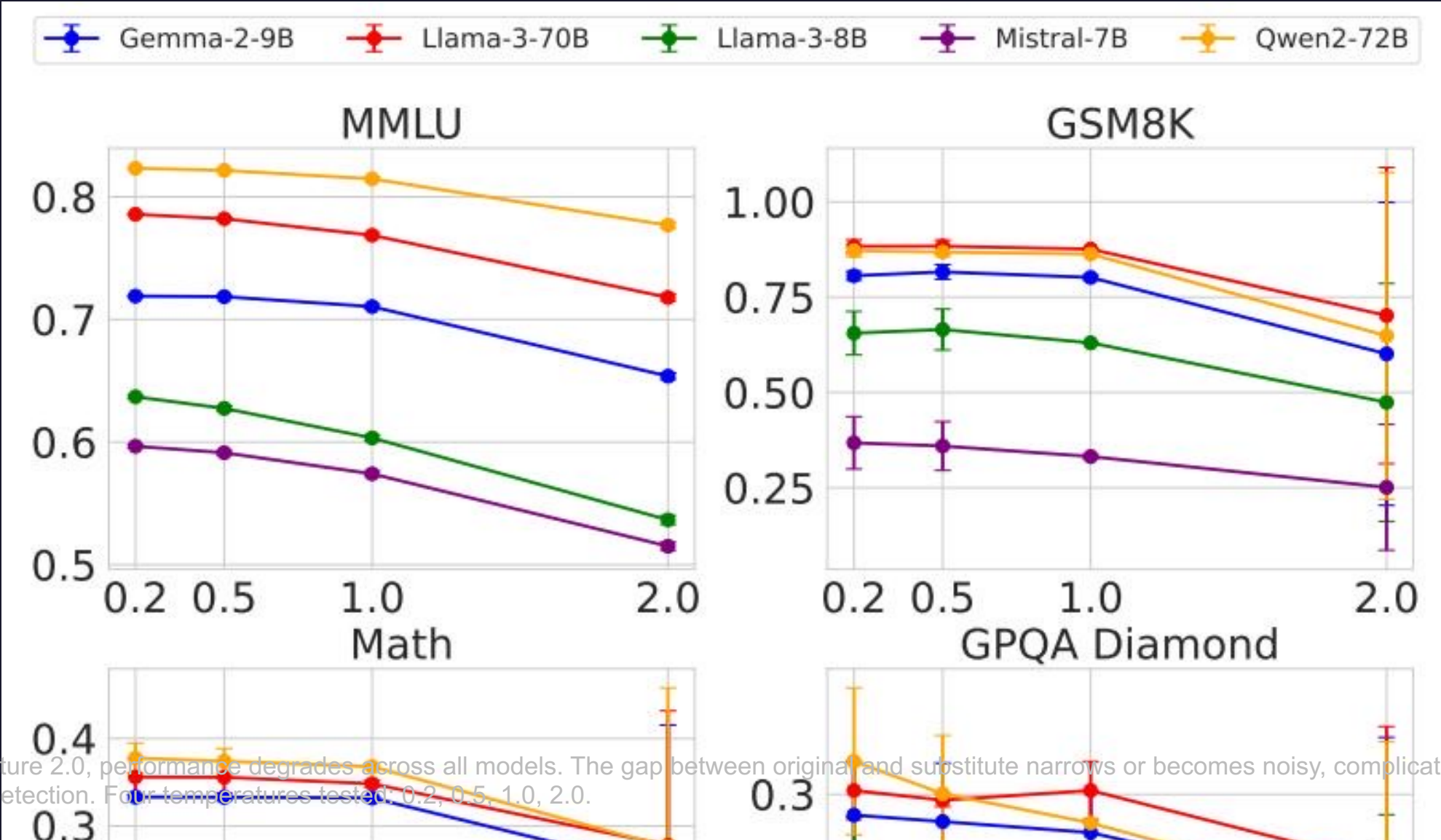
Making Substitution Hard to Detect — Performance Drops Often Within Error Bars

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Note: Green rows = Original precision | Orange rows = FP8 quantized variant

Higher Temperature Amplifies Performance Gaps but Increases Noise

figure_1.jpg — Benchmark Accuracy vs Temperature (5 Models × 4 Tasks)



At temperature 2.0, performance degrades across all models. The gap between original and substitute narrows or becomes noisy, complicating statistical detection. Four temperatures tested: 0.2, 0.5, 1.0, 2.0.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

figure_5.jpg — Log Probability Variation Across Inference Frameworks & Hardware

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Same Llama-3-8B model produces different log probabilities across:

- Software: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- Hardware: NVIDIA A100 vs H100

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Provider mixes substitute responses with probability p , routing only a fraction of queries to the cheaper model. As $p \rightarrow 0$, the mixed distribution P_{mixed} converges to $P_{\text{spec}}(y|x)$, the specified model's distribution. The auditor's samples become statistically indistinguishable from genuine responses at low substitution rates. Detection would require exponentially more queries as p approaches zero — fundamentally query-prohibitive. Adversarial countermeasures include benchmark evasion: known benchmark prompts are routed to the genuine model while substitutes are served for regular traffic.

$$P_{\text{mixed}}(y|x) = (1-p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x) \rightarrow \text{converges to } P_{\text{spec}} \text{ as } p \rightarrow 0$$

Software-Only Auditing Methods Are Fundamentally Unreliable

Three Failure Modes

Method Type	Weakness	Practical Limitation	
Text-Based Statistical Tests (e.g. MMD on output distributions)	Requires impractically large query budgets; fails against subtle or randomized substitutions	Statistical power collapses as substitution probability $p \rightarrow 0$; benchmarks need thousands of samples	
Log-Probability Fingerprinting (token-level metadata comparison)	Defeated by production inference nondeterminism across software versions and GPU hardware	Cannot distinguish genuine model on different infra vs. actual substitution; metadata unreliable in production	
Adversarial Countermeasures (benchmark evasion + randomized routing)	Provider can actively route known benchmarks to genuine model; mixed distributions mask substitution	Software auditors cannot detect routing logic; detection becomes query-prohibitive at low substitution rates	
Benchmark Comparisons (known evals vs. API responses)	Precision loss from FP8/INT8 quantization falls within error bars on MMLU, GSM8K	Auditor cannot attribute small score differences to quantization vs. random noise; FP8 drops are < 2% on most benchmarks	

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware Attestation as the Actionable Path to Verifiable LLM Inference

Hardware-secured enclaves provide cryptographic attestation of code and data executed in the API backend. Attestation proves that specific model weights are loaded and that correct inference code is running. Only modest performance overhead — practical to deploy today (e.g. NVIDIA confidential computing on H100 GPUs).

Unlike ZKPs which are computationally prohibitive for large models, TEEs are deployable at production scale. TEE properties: Provable (cryptographic guarantee) | Efficient (modest overhead) | Robust (hardware-level, not software-level circumvention possible)

Contrast: Software methods can be circumvented; TEE guarantees hold at the hardware level.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and practically undetectable via software alone.
2. FP8 quantization barely affects benchmarks — making it an attractive but nearly undetectable substitution vector.
3. Inference nondeterminism across software versions and GPU hardware kills log-probability fingerprinting.
4. Randomized substitution and benchmark evasion defeat all statistical detection approaches at low substitution rates.
5. Trusted Execution Environments (TEEs) are the only currently viable path to verifiable, auditable LLM APIs.

Call to Action for the Community

Adopt hardware attestation standards for LLM APIs. Demand TEE-based inference from providers. Build open attestation tooling to enable independent verification of model integrity.